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Non-Convex AC/DC Static Transmission Expansion Planning Considering Energy Storage Systems and Uncertainty

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***Abstract*—The integration of non-dispatchable renewable energy sources increases demands for the inclusion of modern technologies, such as High Voltage Direct Current (HVDC) links and Energy Storage Systems (ESS), in the network expansion planning. Also, the current and future power network complexity demands more accurate models to provide more realistic solutions for the transmission expansion planning problem. However, there are no approaches that allow the selection of the proper technology in a unified and accurate TNEP formulation. This article proposes a non-convex formulation to solve the problem, using AC/DC transmission links and ESS. The Iterated Greedy (IG) algorithm is employed as an optimization technique to find good quality solutions. Since uncertainty in load and generation scenarios significantly impacts the Transmission Network Expansion Planning (TNEP), the Monte Carlo Simulation (MCS) technique is used to deal with their stochastic behavior. The advantages of the proposed approach are illustrated using a modified 6-bus Garver test system.**

***Index Terms*—AC/DC network, Energy Storage Systems, Heuristics, Monte Carlo Simulation, Renewable Energy, Transmission Expansion Planning, Uncertainty**

I. Introduction

The Transmission Network Expansion Planning (TNEP) problem aims to determine the least cost set of investments in the transmission infrastructure necessary to meet the balance between demand and generation within a planning horizon. The static TNEP problem considers a single target year, while in the dynamic TNEP problem, the planning horizon is divided into several stages [1]. Looking several decades ahead, using electricity storage and flexible transmission technologies such as High Voltage Direct Current (HVDC) technology is inevitable in the presence of large shares of renewable generation sources.

HVDC systems are currently a technically and economically feasible for bulk power transmission over long distances, connection of non-synchronous networks and weak grids, and submarine and underground transmission. Furthermore, it does not present stability problems when transmitting over long distances compared to classical AC transmission. In [2], a robust multistage TNEP linear model of hybrid high-voltage AC-DC networks is proposed. A hybrid AC/DC TNEP algorithm using a stochastic simulation method and a mixed-integer linear formulation (MILP) is presented in [3].

Energy storage systems (ESS) are an attractive option to increase the flexibility of electric networks. The energy is stored in case of low electricity prices or excessive generation and is returned into the grid during high prices or low available generation. Most research works using ESS deal with the TNEP using a DC network representation without considering other technologies [4], [5]. Those approaches do not allow the proper cost-effective technologies and topology to be chosen for the transmission system's needs. Also, previous research works use linear optimization models, which provide non-realistic solutions for the current and future electricity networks. In this research work, a more accurate and unified non-convex formulation to solve the TNEP problem, using AC/DC transmission links and ESS, is proposed. The growing integration of renewable energy sources has intensified the unpredictability of the net power injections at individual nodes. Consequently, planning considering uncertainty has garnered renewed attention, aiming to create cost-effective and resilient designs that can accommodate all foreseeable scenarios of net power injections. This paper used the Monte Carlo Simulation (MS) technique to deal with the uncertainty and the IG technique to solve the optimization problem.

The paper is organized as follows: Section II presents the formulation and mathematical model. The optimization approach based on an iterated greedy algorithm is presented in Section III. In Section IV, results are presented. The conclusions are summarized in Section V.

II. Formulation

A. AC Power Flow

The nodal active and reactive power flows in the AC network are represented in (1) - (2) as follows:

$$P(V,\theta) = V_i \cdot \sum_{j \in \Lambda^{ac}} V_j \cdot (G_{ij} \cdot \cos\theta_{ij} + B_{ij} \cdot \sin\theta_{ij}) \quad (1)$$

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$$Q(V,\theta) = V_i \cdot \sum_{j \in \Lambda^{ac}} V_j \cdot (G_{ij} \cdot \sin\theta_{ij} - B_{ij} \cdot \cos\theta_{ij}) \quad (2)$$

where V is the nodal voltage, G is the series conductance, and B is the series susceptance, for $i-j$ branches.

B. Mathematical formulation (STNEP)

This section explains the mathematical model and some important aspects related to the formulation. The problem is divided into two sub-problems: *i)* the master transmission expansion problem and *ii)* the operational problem (solved by a multiperiod AC/DC-OPF formulation) [6]. The expansion master problem allows for minimizing the investment and operational costs in the system and it is represented as follows:

$$\textbf{min } v = \mathbb{E}[c_{exp} + w] \quad (3)$$

$$c_{exp} = \sum_{(k,l)\in\Omega} (c_{kl} \cdot n_{kl}) + \sum_{(i,j)\in\Gamma} (c_{dc,ij} \cdot n_{dc,ij}) + \sum_{h\in\Delta} (cc_h) + \sum_{y\in\Lambda} \sum_{(t=1)}^{T} (c_{F,y} \cdot n_{ess,y} + c_{P,y}(P^t_{c,y} + P^t_{d,y}) + c_{E,y} \cdot e^t_y) \quad (4)$$

s.t.

$$0 \le n_{kl} \le \overline{n_{kl}}; \quad n_{kl} \in \mathbb{Z} \quad (5)$$

$$n_{dc,ij} \le n_{dc,ij} \le \overline{n_{dc,ij}}; \quad n_{dc,ij} \in \mathbb{Z} \quad (6)$$

where $\mathbb{E}$ represents the expected value associated to the objective function (3). c_{kl}, c_{dc}, cc_h are the costs of candidate AC lines, DC lines, and converters, respectively. c_F, c_P, c_E are the fixed and power and energy-dependent variable costs of the ESS, respectively. Ω, Γ, Δ, and Λ are the sets of candidate AC circuits, DC circuits, converters, and nodes for ESS, respectively. P^t_c, P^t_d, and e^t represent the charge and discharge power and the energy stored in time t, respectively. n_{kl}, n_{dc}, n_{ess} are the number of added elements corresponding to the AC circuits, DC circuits, and converters. Equations (5), (6) represent the limits of AC circuits and DC circuits added with a maximum of five elements used in this paper. For ESS, only one unit is allowed to be added per bus. The scalar w represents simultaneously the load-shedding cost for the operational problem and a penalty strategy for the master objective function. The operational problem allows determining the load shedding for active and reactive power in the topology (7) through a multiperiod AC optimal power flow formulation with operational constraints (8) - (23). The active and reactive load shedding is modeled by fictitious generators (L_P, L_Q), connected to the load nodes, that supply the load when the generation and transmission systems are not able to satisfy the total demand. The purpose of the fictitious generators is to penalize the objective function to discriminate transmission topologies with unattended demand so that the operational optimization problem minimizes the fictitious generation [7]. The objective function of the operational problem (7) is presented as follows.

$$\textbf{min } w = \sum_{(k\in\wedge\cup\Gamma)} (\alpha_1 \cdot L_{Pk}) + \sum_{(k\in\wedge)} (\alpha_2 \cdot L_{Qk}) \quad (7)$$

s.t.

AC Lines

$$P(V,\theta)^t + P^t_{Tf} - U(P^t_G) - P^t_d + P^t_c + H(P^t_D) - L^t_P = 0 \quad (8)$$

$$Q(V,\theta)^t + Q^t_{Tf} - U(Q^t_G) + H(Q^t_D) - L^t_Q = 0 \quad (9)$$

$$\underline{U(P_G)} \le U(P^t_G) \le \overline{U(P_G)} \quad (10)$$

$$\underline{U(Q_G)} \le U(Q^t_G) \le \overline{U(Q_G)} \quad (11)$$

$$\underline{L_P} \le L^t_P \le \overline{L_P} \quad (12)$$

$$\underline{L_Q} \le L^t_Q \le \overline{L_Q} \quad (13)$$

$$S^{fr,t} \le \overline{S} \quad (14)$$

$$S^{to,t} \le \overline{S} \quad (15)$$

Energy Storage Systems

$$e^t = e^{t-1} + (\eta_c \cdot P^t_c - \tfrac{P^t_d}{\eta_d}) \cdot t \quad (16)$$

$$P^t_c \cdot P^t_d = 0 \quad (17)$$

$$0 \le P^t_c \le n_{ess} \cdot \overline{P_{ess}} \quad (18)$$

$$0 \le P^t_d \le n_{ess} \cdot \overline{P_{ess}} \quad (19)$$

DC Lines

$$P^t_{dc} = p \cdot V^t_{dc} \cdot I^t_{dc} \quad (20)$$

$$P^t_{dc,loss} = p \cdot r^t_{dc} \cdot (I^2_{dc})^t \quad (21)$$

$$P^t_{cv,dc} + P^t_{dc} + H(P^t_D) = 0 \quad (22)$$

$$P^t_{dc} \le \overline{P_{dc}} \quad (23)$$

AC/DC Converters

$$P^t_{cv,ac,h} + P^t_{cv,dc,h} = a_{cv} + b_{cv} \cdot I^t_{cv,ac} + c_{cv}(I^t_{cv,ac})^2 \quad (24)$$

$$P^t_{cv,ac} = cos\phi^t_h \cdot S_{cv,ac,rated} \quad (25)$$

$$Q^t_{cv,ac} = sin\phi^t_h \cdot S_{cv,ac,rated} \quad (26)$$

$$I^t_{cv,ac} \le \overline{I_{cv}} \quad (27)$$

$$\underline{V_{cv,h}} \le V^t_{cv,ac,h} \le \overline{V_{cv,h}} \quad (28)$$

Equations (8) and (9) represent active and reactive power balance at each bus, respectively. The power (U) from non-dispatchable renewable sources and load fluctuations (H) are stochastic variables handled by probability density functions. Equations (10) and (11) are the active and reactive generation limits, respectively. The limits of active and reactive fictitious generators are represented by (12) - (13). For the active power, the limits are fixed on 0 and 1000 MW, while reactive power is not considered in this work. Equations (14) and (15) are the capacity of each transmission line. The energy stored in the ESS at time t (e^t), depending on the charge and discharge efficiencies (η_c, η_d), the current power and the energy stored in the previous period ($t-1$), is calculated by (16). Equation (17) restricts the charge and discharge power, ie, the unit can either only charge or only discharge at time t. The limits of the ESS power output between 0 and the rated value (P_{ess}) are represented by (18) - (19). The elements of vectors $P(V,\theta)$ and $P(V,\theta)$ appear in (1) - (2). The DC equations (20) - (23) depends on the DC voltage and current, number of poles ($p = 2$ for this work), and the resistance (r_{dc}). For

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AC/DC converters, equation (24) represents the balance on the AC and DC sides as a function of equipment, switching, and conduction losses (a_{cv}, b_{cv}, c_{cv}). Equations (25)- (26) represent the active and reactive power on the AC side, as a function of the rated power of the converter and the firing angle of the thyristors (ϕ_h). Finally, equations (27)- (28) are the limits of the current and voltage of the converters respectively.

1) Uncertainty: To deal with the uncertainties due to the use of wind power (U) and load fluctuations (H), the probability density functions (PDF) (29)-(30) are used in (8)-(9).

$$d(p) = \frac{1}{B(\alpha, \beta)} \cdot x^{\alpha-1}(1-p)^{\beta-1} \tag{29}$$

$$v = \frac{k}{c}(\frac{p}{c})^{k-1} \cdot e^{(-p/c)^k} \tag{30}$$

For the load model (H), the random data are represented by a beta distribution function (29) that depends on the shape parameters α and β. For the wind power plant, the power generation (U) is modeled by a power output curve [8], and the wind speed is modeled by a Weibull function (30), which depends on the scale c and the shape k parameters.

2) Objective function penalization and load shedding strategy: In this work, the load-shedding concept is used as a penalization strategy to obtain zero load-shedding transmission plans. Reactive load shedding is not considered, and α_2 is set to zero. The load-shedding strategy is thoroughly explained in [7].

3) Energy Storage Systems (ESS): Costs for energy storage systems depend on the power (kW) and energy capacity (kWh) as well as on fixed costs (3) that are independent of the system size. Fixed costs include, for example, control systems, construction management, and permitting. Further information about storage capacity and costs for each system can be found in [9], [10].

III. Iterated Greedy Algorithm

The proposed optimization approach consists of two steps:

1. Initial solution (s): The so-called *Forward* [11] heuristic starts from zero added lines. It consists of adding, one by one, all the candidate transmission circuits (l_i) to the system. For each added circuit, an AC optimal power flow (8) - (23) is performed to verify the feasibility of the current topology, and the resulting w value is passed to (3). This is performed until the first feasible topology ($w = 0$) is found.

2. Optimization algorithm: *Iterated Greedy* (IG) is a metaheuristic that generates a sequence of solutions using two main steps: destruction and (re-)construction. During the destruction step, some solution components are removed from the currently best solution. Then, the *Forward* algorithm is applied to obtain a new candidate solution. If this new solution is better than the currently best one, the new solution is taken as the new currently best solution. The algorithm stops after m consecutive non-improving iterations and returns the best solution found. In this work, it is set to $m = 6$. Figure 1 explains the process.

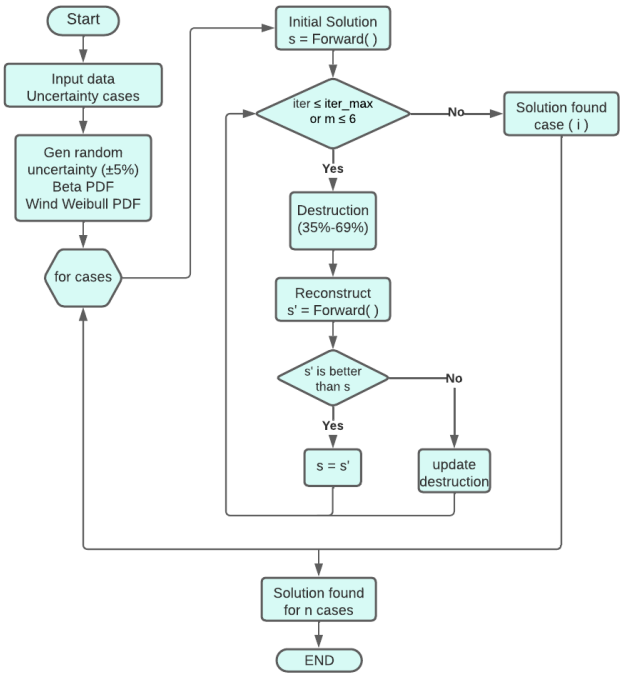


Fig. 1. Flow chart of the optimization algorithm

IV. Results

The modeling layer JuMP [12] in Julia [13] was used to develop the optimization model. The interior point solver Ipopt [14] was used to solve AC/DC-OPF. In addition, a modified Garver 6-bus test system was used for the TNEP tests. The maximum number of allowed circuits per right of way is five. The maximum number of allowed ESS is one. The results were obtained using Julia parallel processing with 16 processors on an AMD Ryzen 9, 3.40GHz, 64GB RAM PC. The demand and wind generation data are distributed within a ±5% of uncertainty range. Simulations are performed using two scenarios. The first scenario compares the AC/DC base system with and without ESS. Also, a robustness analysis of the optimization algorithm is performed using the scenario with ESS. The second scenario corresponds to the TNEP considering storage and uncertainty using 500 Monte Carlo simulations. Table I summarizes the characteristics of the modified Garver system. Costs are in units of (USD $\times 10^6$).

A. AC/DC Garver 6-bus system

This system has 6 buses with a total power demand of 760 MW and 152 Mvar, 6 transmission lines, and an installed capacity of 1,110 MW. For this system, 1 AC candidate link, 11 DC candidate links, and 6 ESSs candidates (one per bus for this paper) are considered for expansion. In subsection 1, dispatchable generation is considered. In subsection 2, dispatchable generator 1 is replaced by a wind generator described in Table I.

1) ESS and algorithm robustness analysis: Table II shows the solution vectors for scenarios with and without ESS.

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TABLE I
MODIFIED GARVER SYSTEM DATA

Cost Lines			Gen	
	Line	Cost	Bus	MW
AC	5-6	10	1	150
	2-5	5.3	3	360
	2-6	5.4	6	600
	2-6	6.4	**Storage**	
	3-6	5.6	Rated Power	50 MW
	3-6	6.6	Fixed Cost	2
DC	4-5	5.7	Power Cost ($/kW)	500
	4-5	6.7	Capacity Cost ($/kWh)	800
	4-6	5.8	**Uncertainty**	
	4-6	6.8	$\alpha - \beta$	1 - 1
	5-6	5.9	c - k	11.1m/s - 2.16
	5-6	6.9		

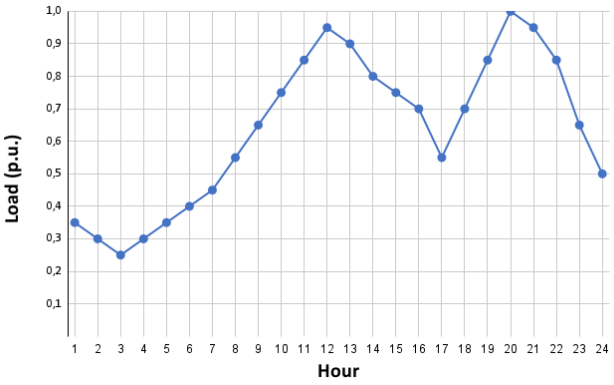


Fig. 2. Load curve

When storage systems are considered, the total costs decrease. Variable costs, as shown in Section II, depend on the transferred power and energy stored in the ESS; these costs represent about 15% of the total cost. Therefore, savings can be achieved when storage systems are considered an option for expansion in the TNEP problem. The computing time increases considerably since ESS are considered, and a multi-period optimal flow calculation is carried out according to the load curve shown in Figure 2.

TABLE II
COMPARISON OF THE GARVER SYSTEM WITH AND WITHOUT STORAGE. WITHOUT UNCERTAINTY

Additional lines	AC/DC Garver 6-bus without uncertainty	
	Without Storage	With Storage
AC links		
5 - 6	1	1
DC links (cost)		
2 - 6	1	-
4 - 6	1	1
ESS (bus)		
2	-	1
Time (seg)	76 s	290 s
Fixed Cost (USD $\times 10^6$)	43.7	33.8
Variable Cost (USD $\times 10^6$)	-	5.57
Total Cost (USD $\times 10^6$)	43.7	39.37

To show the algorithm's robustness, the scenario considering ESS was selected, where the load and generation vary according to the percentages shown in Table III. Results prove that

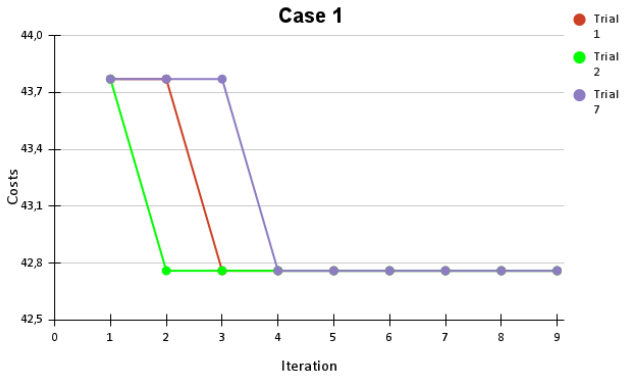


Fig. 3. Convergence for Scenarios 1 and 2 with uncertainty

for 10 trials, the same solution was obtained, which is shown in Table III. The 100% success in obtaining the same solution demonstrates the algorithm's robustness.

TABLE III
RESULTS. 10 TRIALS. AVERAGE VALUES

	Variation
	ΔPD(1): 3.1%, ΔPD(2): -3.3%, ΔPD(3): 2.2% ΔPD(4): 2.7%, ΔPD(5): -0.5%; ΔPG(1): -2.6%
DC links	
4 - 6	1
5 - 6	1
ESS (bus)	
2	1
Success	100%
Fixed Cost (USD $\times 10^6$)	37.2
Variable Cost (USD $\times 10^6$)	5.56
Total Cost (USD $\times 10^6$)	42.76

Figure 3 shows the convergence of the heuristic algorithm for some trials. It is important to highlight that the final solution has already been obtained from the fourth iteration. The algorithm stops after a few more iterations since there is no improvement in the objective function.

2) TNEP with uncertainty: When working with uncertainty, each load is considered a variable under a Beta Distribution. Also, the wind speed is considered a stochastic variable with values provided by the Weibull Distribution. Table IV summarizes the robust plan obtained by performing 500 MCS. Most of the topologies for the robust plan are the same. The variable cost is different due to the uncertainty. The table shows an average value for the costs and computing time.

Figure 4 represents the global solutions obtained for the 500 MS. The expected plan, shown in Table IV, is considered a robust plan. This topology is capable of dealing with most uncertain scenarios. The convergence in the scenarios, including uncertainty, is similar to that shown in Figure 3.

Finally, Figure 5 summarizes the behavior of the ESS throughout the 24 time periods of the load curve (Figure 2). ESS delivers energy around the 12th hour when there is peak demand. Then, ESS is charged during low-demand periods to deliver energy around the 20th hour.

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TABLE IV
TNEP WITH UNCERTAINTY. 500 CASES.

Aditional Lines	Robust Plan
DC links	
4 - 6	1
5 - 6	1
ESS (bus)	
2	1
Time (seg)	297.60 seg
Fixed Cost (USD $\times 10^6$)	37.2
Variable Cost (USD $\times 10^6$)	5.62
Total Cost (USD $\times 10^6$)	42.82

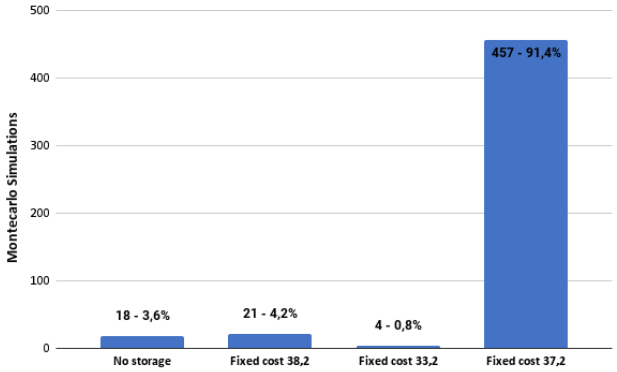


Fig. 4. Solutions obtained with uncertainty

V. CONCLUSIONS

The TNEP is a very complex problem, and it must be rethought when new technologies arise. ESS devices can help mitigate operational problems in the electric power system, lowering overall costs. On the other hand, renewable energies are booming and should also be considered in the TNEP problem. However, those technology conditions demand the use of more accurate and unified TNEP formulations, including uncertainty. The proposed model considers AC/DC links, ESS, and uncertainty in load and generation.

For the analyzed scenarios, results show that ESS can be a viable option for expansion, where the final cost achieved savings compared to the case without including the ESS in the TNEP problem.

The proposed approach demonstrated great robustness for a simple test network, although the complexity is still very high due to the multiperiod and uncertainty features. The proposed model can be extended for a multistage TNEP problem, considering contingencies.

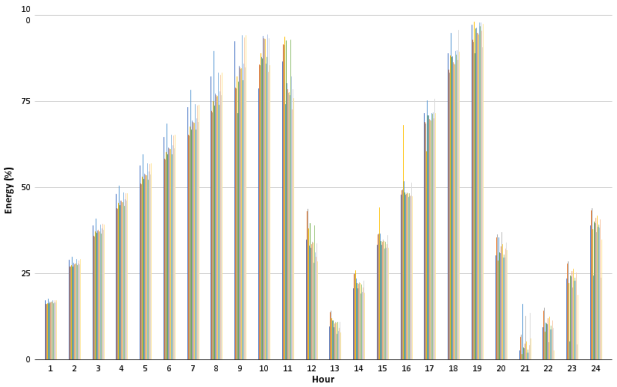


Fig. 5. Storage capacity at each time period

VI. ACKNOWLEDGEMENTS

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